Cluster Analysis Research Report

An Investigation into the StarWars Data

Paul Kieffaber

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Contents

The Star Wars Dataset **1**

Pre-process & feature selection **2**

Decide the number of clusters with elbow & silhouette **4**

Fit and visualize three methods **5**

K-means Clustering (Hartigan-Wong method) 5

Heirarchical Clustering 7

Density Based Clustering (DBSCAN) 8

Pick a final solution & justify **10**

Validate clusters on an external variable **11**

Goal. Pick a publicly available dataset, ask a concrete question about how observations might group together, then (1) determine a sensible number of clusters using the elbow and silhouette methods, (2) compare K-means, Hierarchical (Ward.D2), and DBSCAN, (3) select and justify a final solution, and (4) compare clusters on at least one validation variable not used to form clusters.

The Star Wars Dataset

The starwars dataset (included in the dplyr package) contains information on 87 characters from the Star Wars universe.

- Some Key variables:

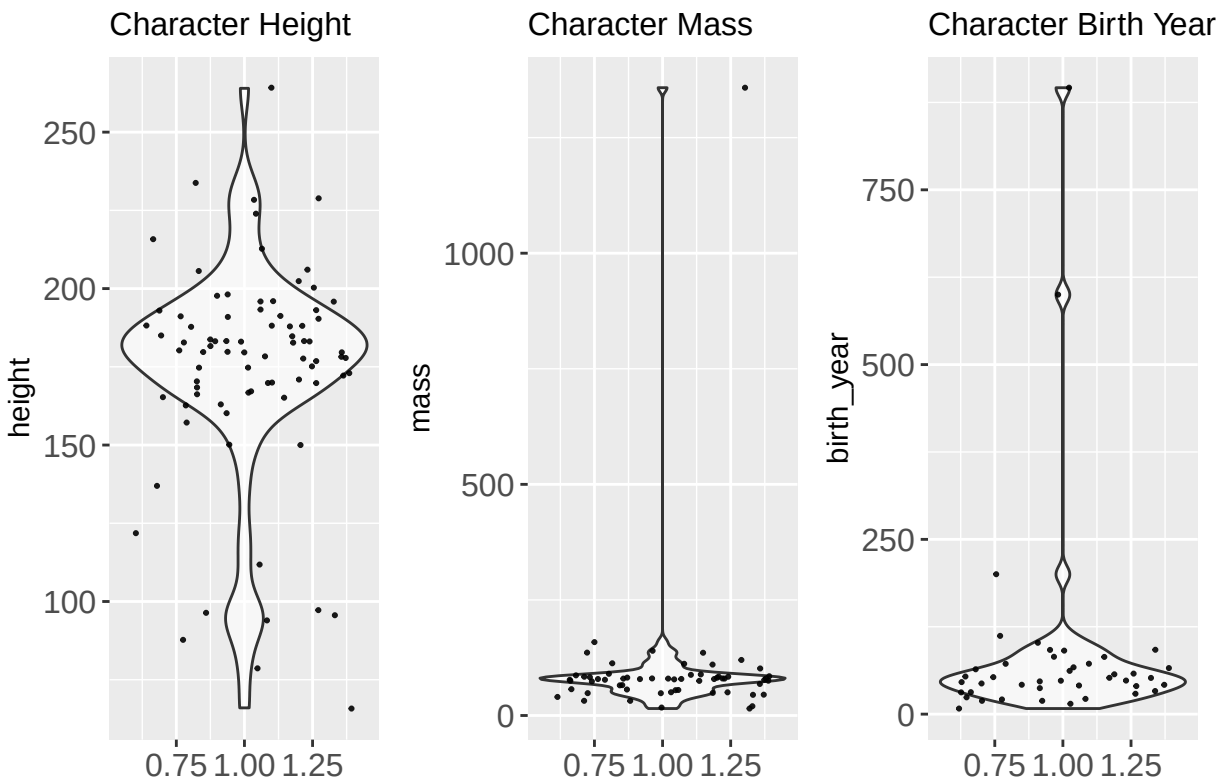
```
## Rows: 87
## Columns: 14
## $ name       <chr> "Luke Skywalker", "C-3PO", "R2-D2", "Darth Vader", "Leia Or~
## $ height     <int> 172, 167, 96, 202, 150, 178, 165, 97, 183, 182, 188, 180, 2~
## $ mass       <dbl> 77.0, 75.0, 32.0, 136.0, 49.0, 120.0, 75.0, 32.0, 84.0, 77.~
## $ hair_color <chr> "blond", NA, NA, "none", "brown", "brown, grey", "brown", N~
## $ skin_color <chr> "fair", "gold", "white, blue", "white", "light", "light", "~
## $ eye_color  <chr> "blue", "yellow", "red", "yellow", "brown", "blue", "blue",~
## $ birth_year <dbl> 19.0, 112.0, 33.0, 41.9, 19.0, 52.0, 47.0, NA, 24.0, 57.0, ~
## $ sex        <chr> "male", "none", "none", "male", "female", "male", "female",~
## $ gender     <chr> "masculine", "masculine", "masculine", "masculine", "femini~
## $ homeworld  <chr> "Tatooine", "Tatooine", "Naboo", "Tatooine", "Alderaan", "T~
## $ species    <chr> "Human", "Droid", "Droid", "Human", "Human", "Human", "Huma~
```

```
## $ films      <list> <"A New Hope", "The Empire Strikes Back", "Return of the J~
## $ vehicles   <list> <"Snowspeeder", "Imperial Speeder Bike">, <>, <>, <>, "Imp~
## $ starships  <list> <"X-wing", "Imperial shuttle">, <>, <>, "TIE Advanced x1",~
```

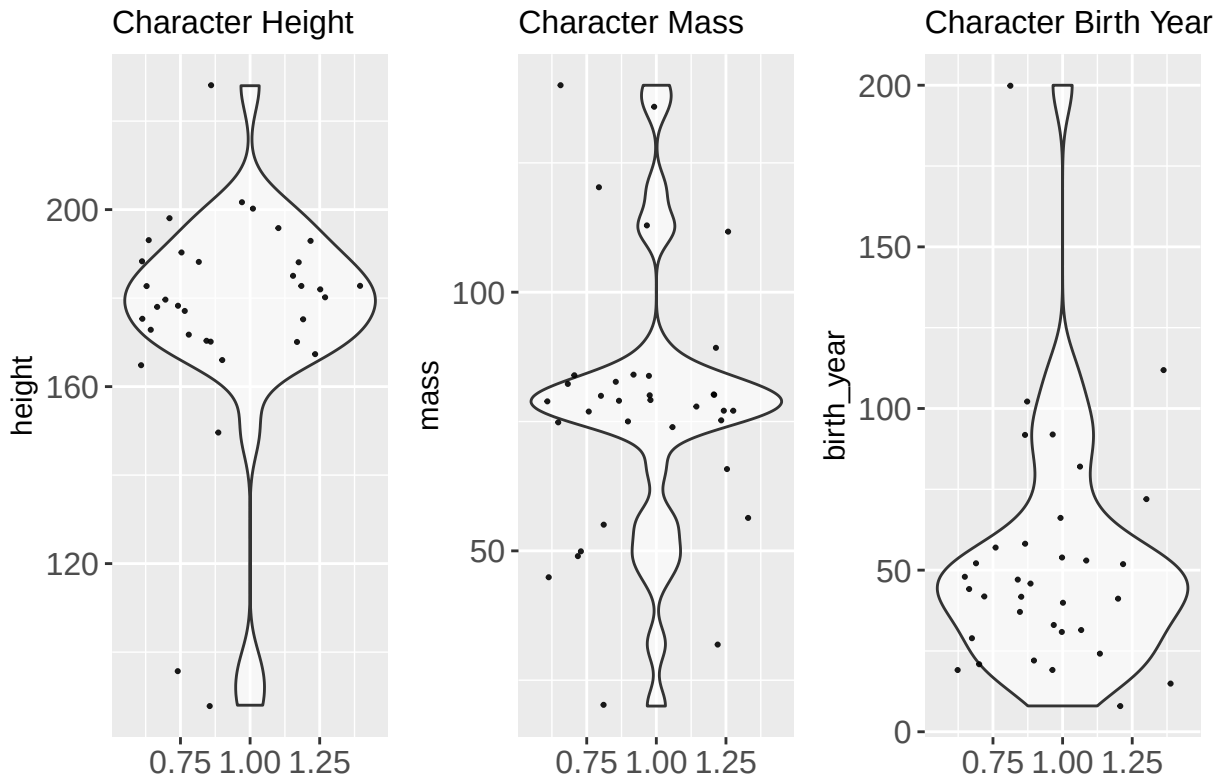
Research Question Are there unique groupings of Star Wars characters based on their physical characteristics?

Pre-process & feature selection

- There are three continuous variables that can be used for clustering:
  - Height
  - Mass
  - Birth_year



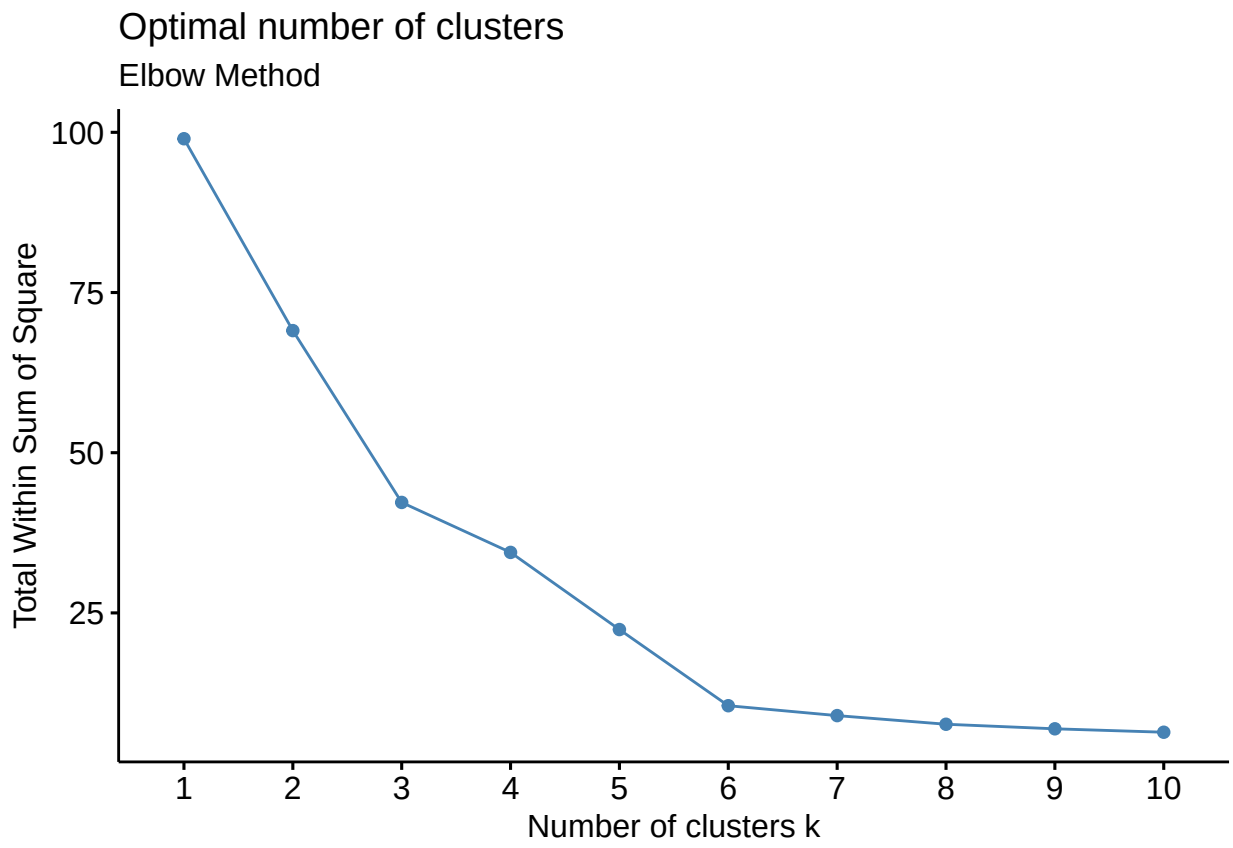
Looking at these variables, it appears that Mass>500 and Birth_year>250 will remove outliers in those variables (see filtered data below).

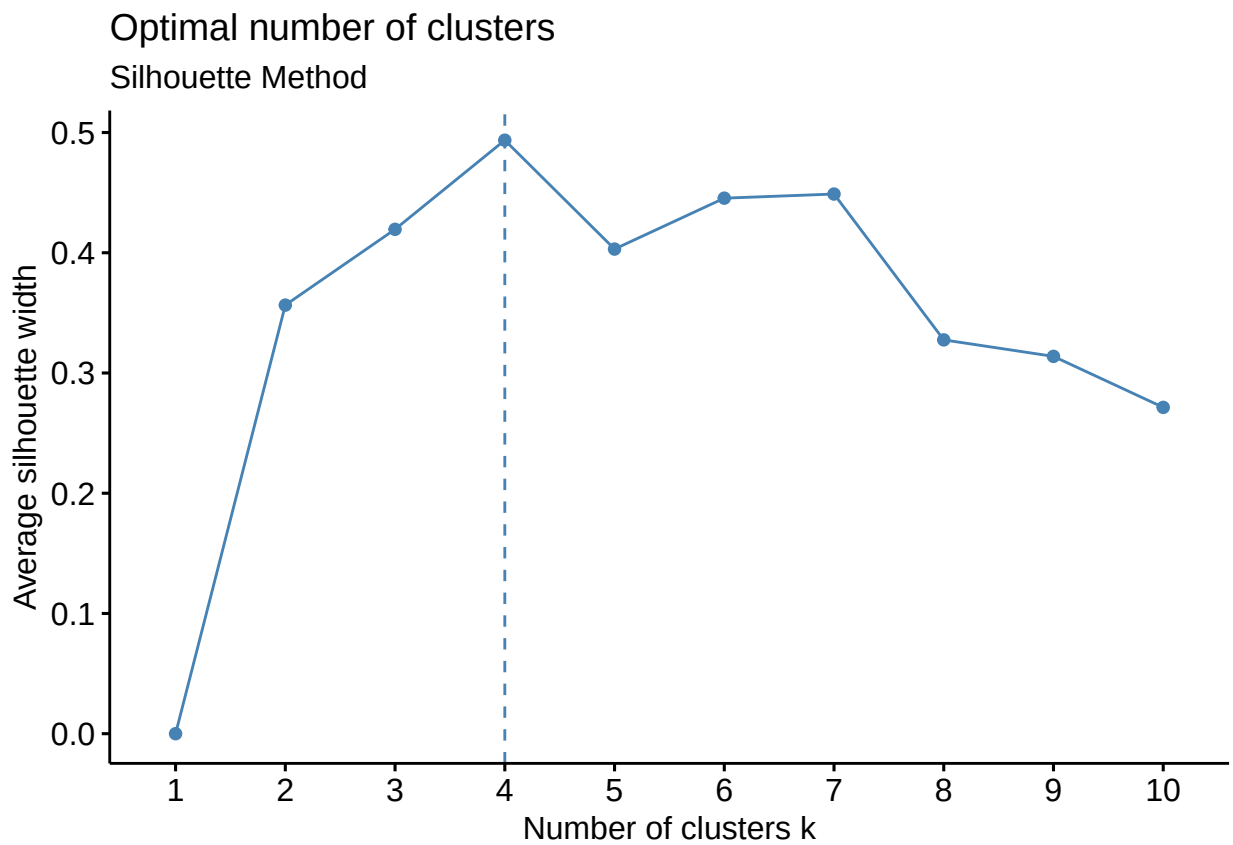


Data for clustering were then normalized (see below).

```
##           height       mass  birth_year
## [1,] -0.18628834 -0.0552317 -0.92159766
## [2,] -0.38177611 -0.1344670  1.64069180
## [3,] -3.15770240 -1.8380271 -0.53587667
## [4,]  0.98663826  2.2822111 -0.29066832
## [5,] -1.04643452 -1.1645266 -0.92159766
## [6,]  0.04829698  1.6483283 -0.01239817
```

Decide the number of clusters with elbow & silhouette



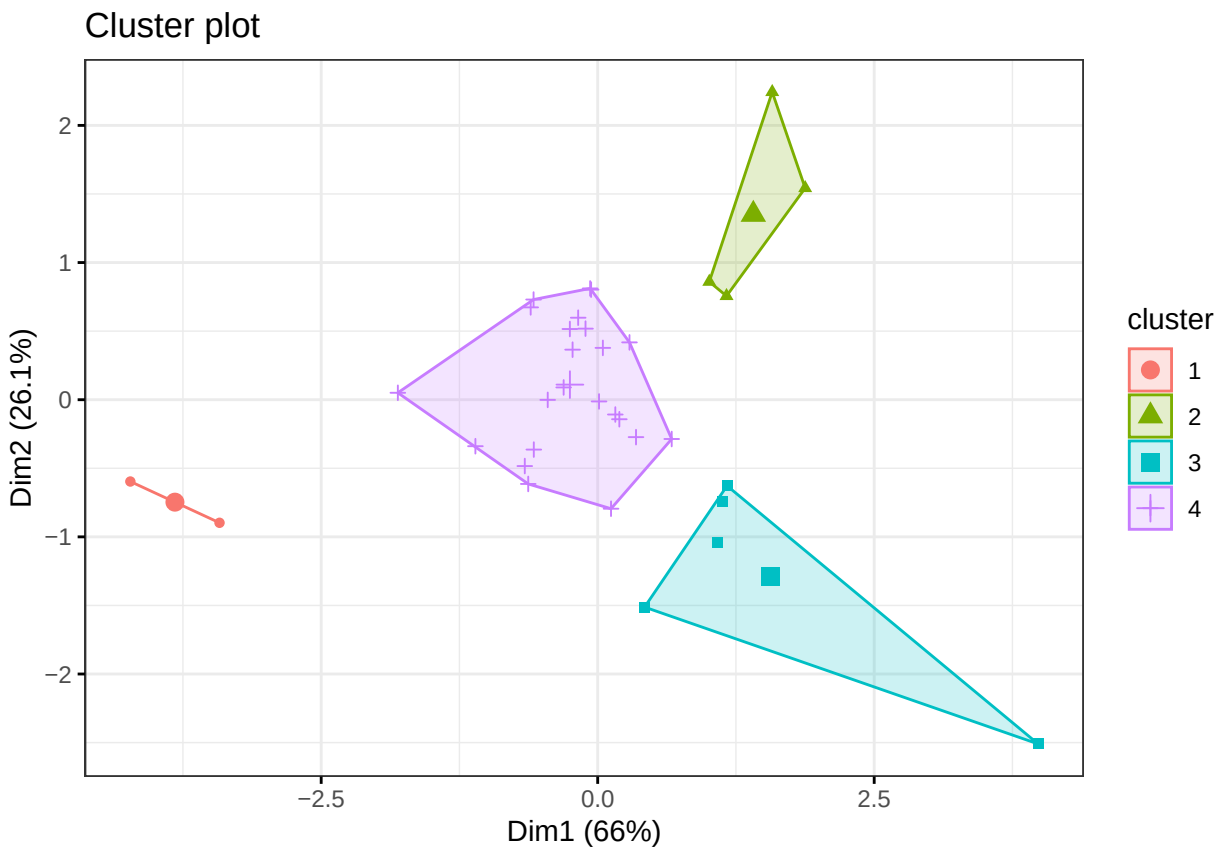


The elbow plot and silhouette method appear to be in agreement that four clusters may be the most appropriate (i.e., $k = 4$).

Fit and visualize three methods

K-means Clustering (Hartigan-Wong method)

The PCA plot shows good separation of the clusters with the two PCA dimensions accounting for more than 92% of the variabiltiy in the data.



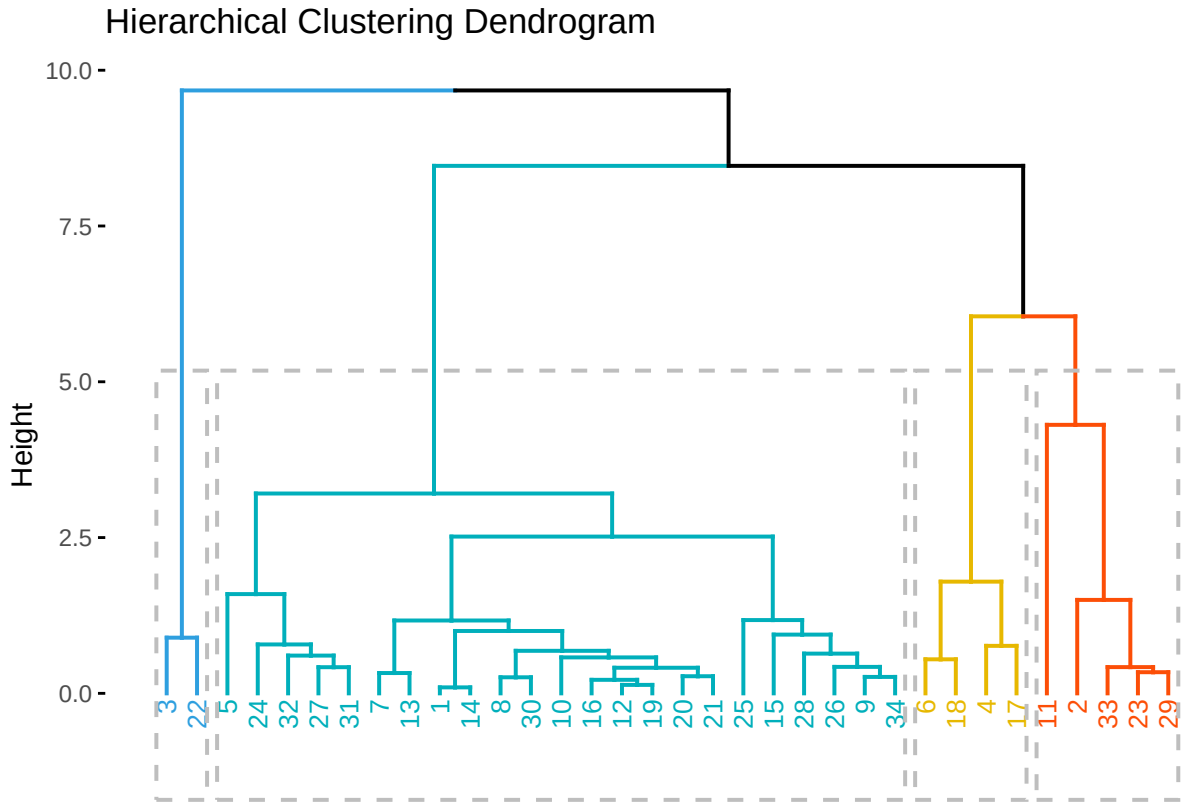
CLUSTER SUMMARY:

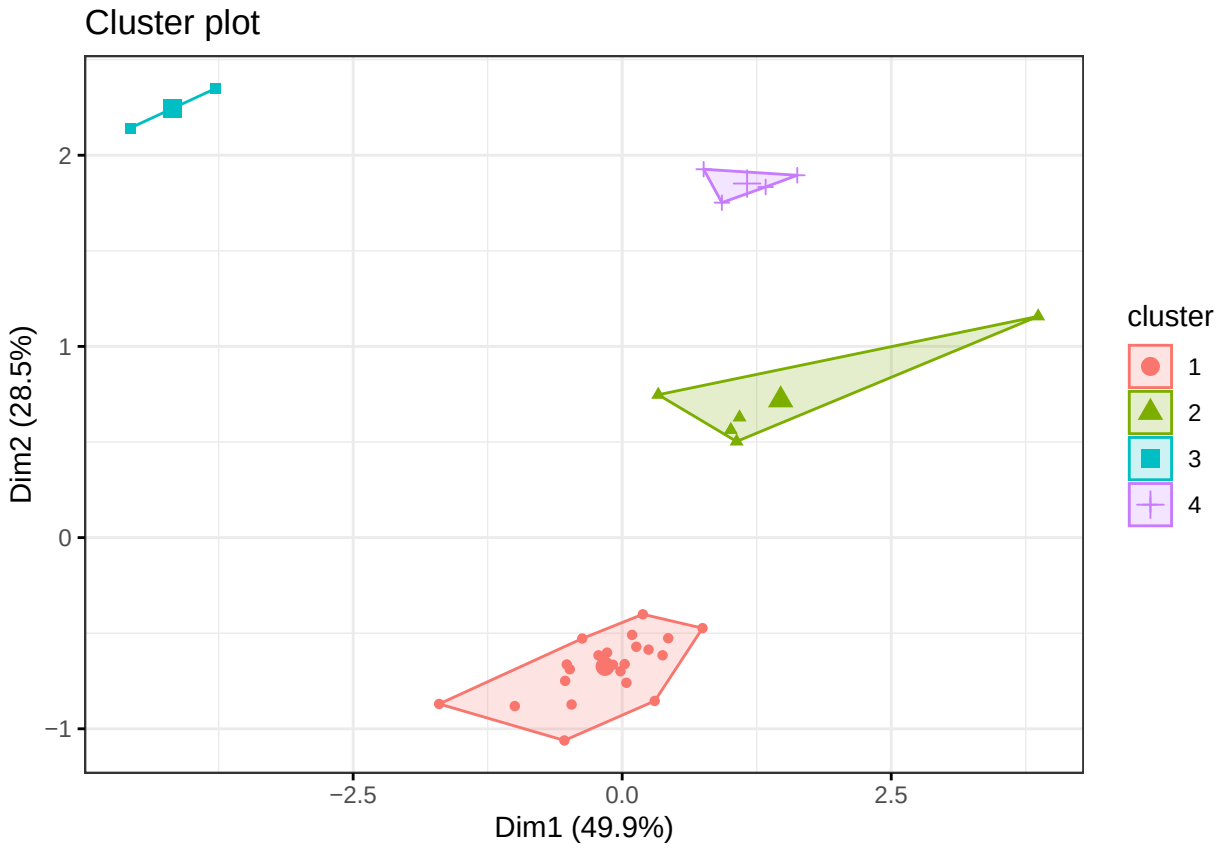
```
## # A tibble: 4 x 5
##   KMcluster     n Height_mean Mass_mean Age_mean
##       <int> <int>       <dbl>     <dbl>    <dbl>
## 1         1     2      -3.31      -2.08   -0.880
## 2         2     4       0.615      1.94   -0.330
## 3         3     5       0.744      0.365   1.85
## 4         4    23       0.0194    -0.235  -0.268
```

It looks like most of my clusters are pretty small. ***Interpretation:***

- Cluster 1: Short, light, and young.
- Cluster 2: Tall, heavy, average age
- Cluster 3: Tall, average mass, very old
- Cluster 4: Average height, average mass, average age

Heirarchical Clustering



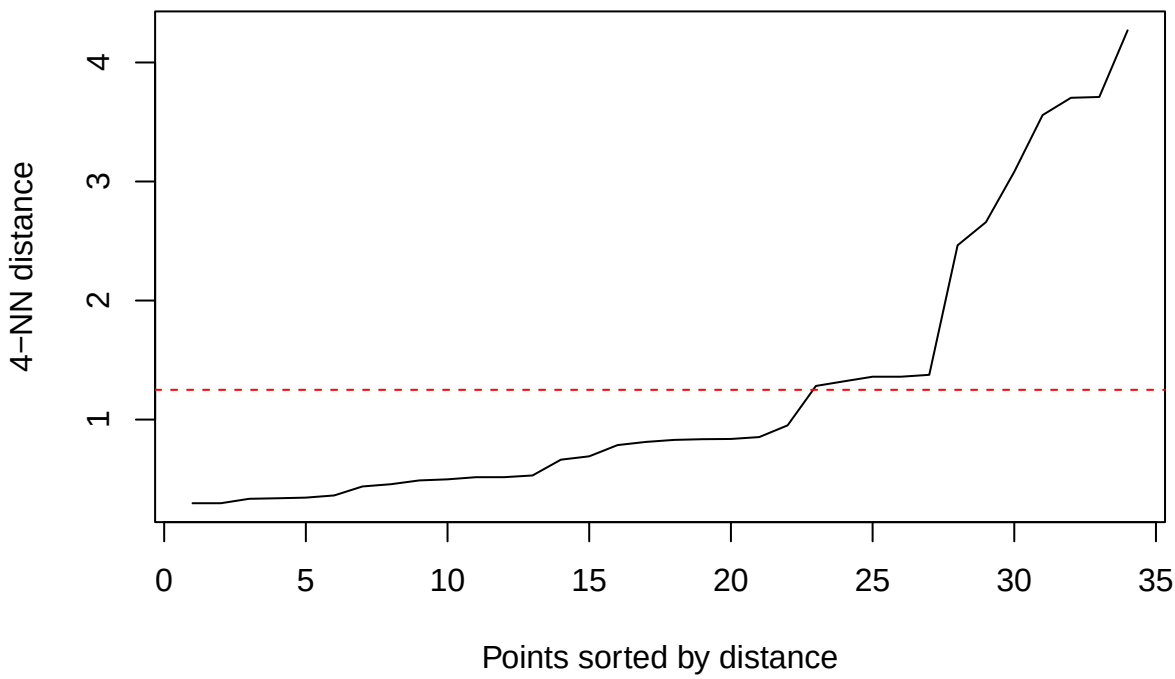


CLUSTER SUMMARY:

```
## # A tibble: 4 x 5
##   KMcluster     n Height_mean Mass_mean Age_mean
##       <int> <int>       <dbl>     <dbl>    <dbl>
## 1         1    23      0.0194    -0.235   -0.268
## 2         2     5      0.744      0.365    1.85
## 3         3     2     -3.31      -2.08    -0.880
## 4         4     4      0.615      1.94    -0.330
```

The clusters identified using hierarchical clustering are very similar to those using K-means clustering.

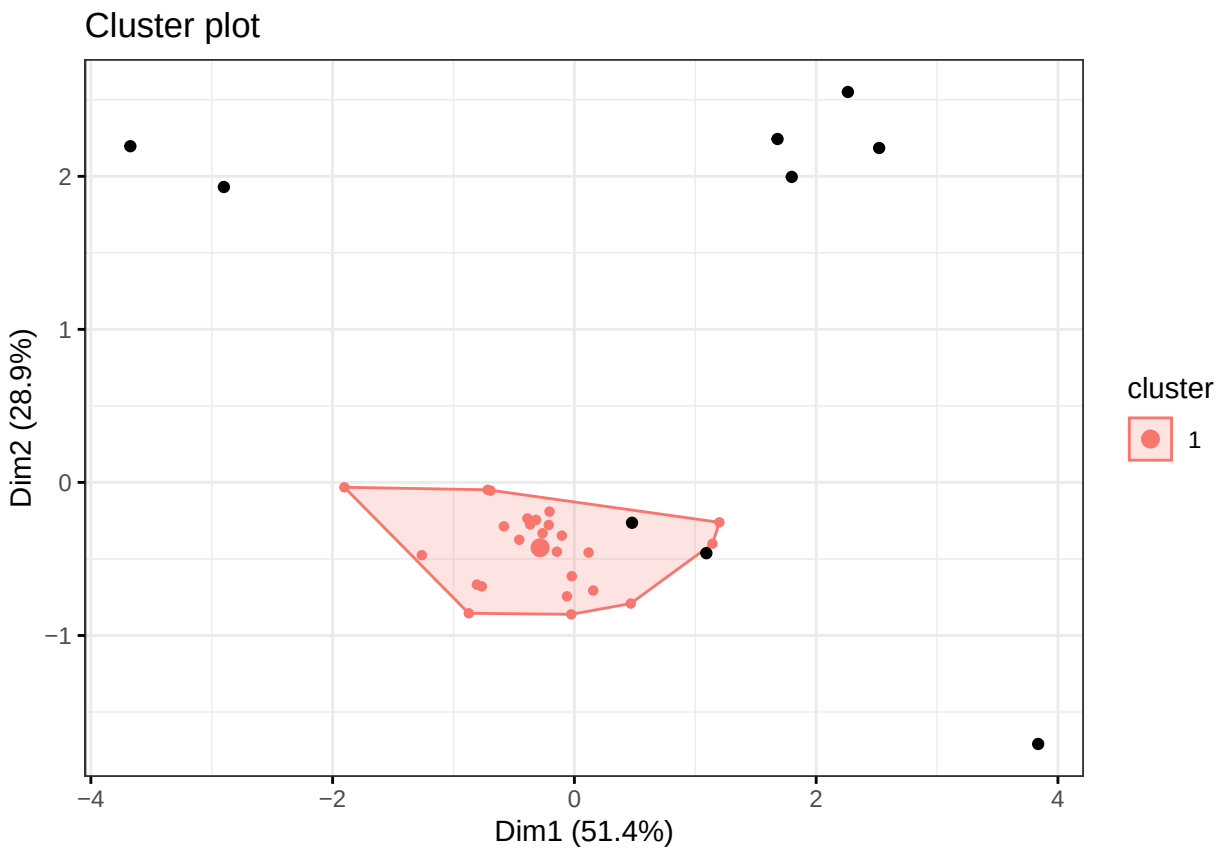
Density Based Clustering (DBSCAN)



- The nearest neighbor distance plot indicates an epsilon of about 1.25 with k=4, so I'll use epsilon=1.24 with minpnts=5 (k+1)

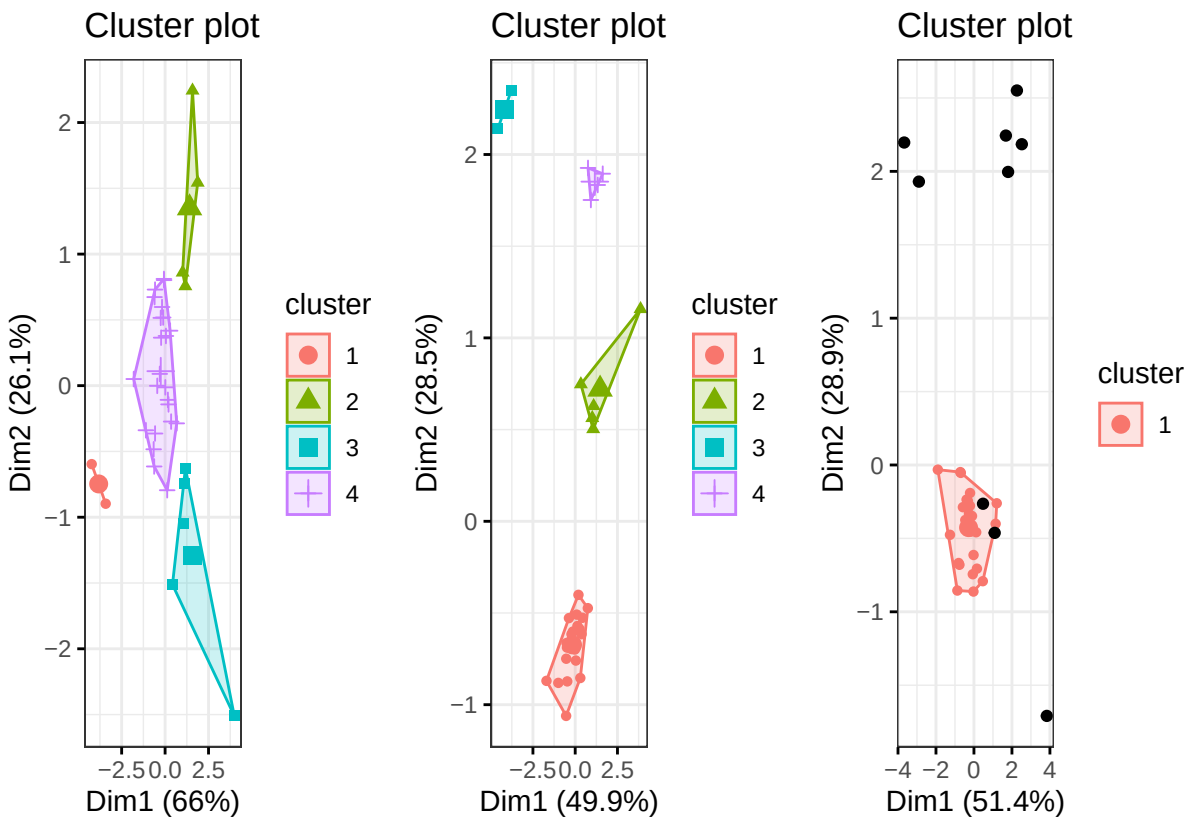
```
## DBSCAN clustering for 34 objects.
## Parameters: eps = 1.25, minPts = 5
## Using euclidean distances and borderpoints = TRUE
## The clustering contains 1 cluster(s) and 9 noise points.
##
##  0  1
##  9 25
##
## Available fields: cluster, eps, minPts, metric, borderPoints
```

Ok, so the DBSCAN method suggests there are no natural clusters, that everyone should be in the same cluster with 9 outliers.



This solution seems strange. When reduced to 2 dimensions, two of the outliers fall within the cluster.

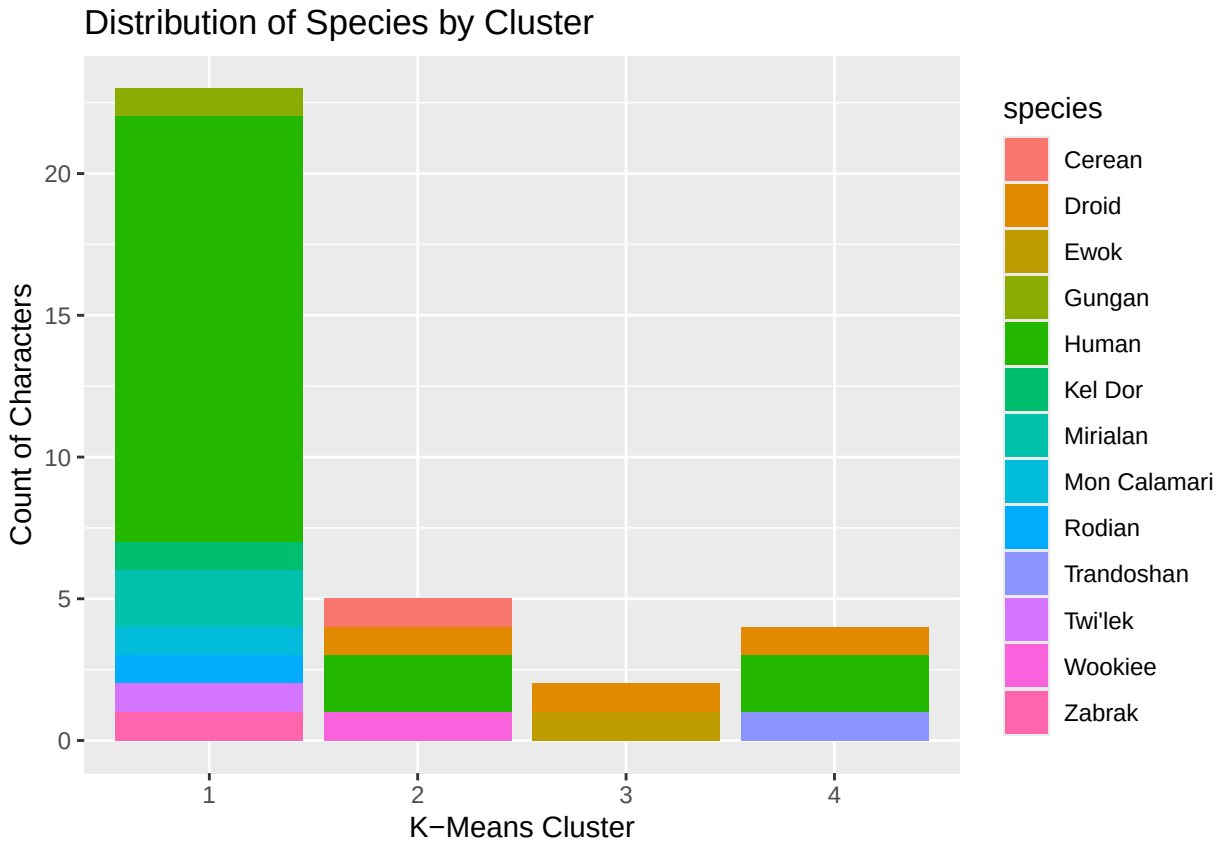
Comparison



Pick a final solution & justify

Kmeans & Hierarchical were nearly identical. So I'm going to pick the k-means for validation

Validate clusters on an external variable



In order to validate the clustering, I decided to test whether certain species were more or less likely to be in a cluster using a chi-squared test of association.

```
##
##  Pearson's Chi-squared test
##
## data:  contingency_table
## X-squared = 48.42, df = 36, p-value = 0.08085
```

The results indicate that the star wars characters are NOT clustered by physical characteristics into distinct species. The Chi-squared test did not reach statistical significance, indicating that clustering on physical attributes did not relate to species.